SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 3

Data-Driven Decision Making, Descriptive Analytics & Storytelling

Decision types and stages, intuition vs data-driven decisions, the data–intuition partnership, DIKW, analytics levels (Zara), descriptive statistics, visualisation and chart choice, data storytelling and decluttering.

Material

1 Decisions and DDDM

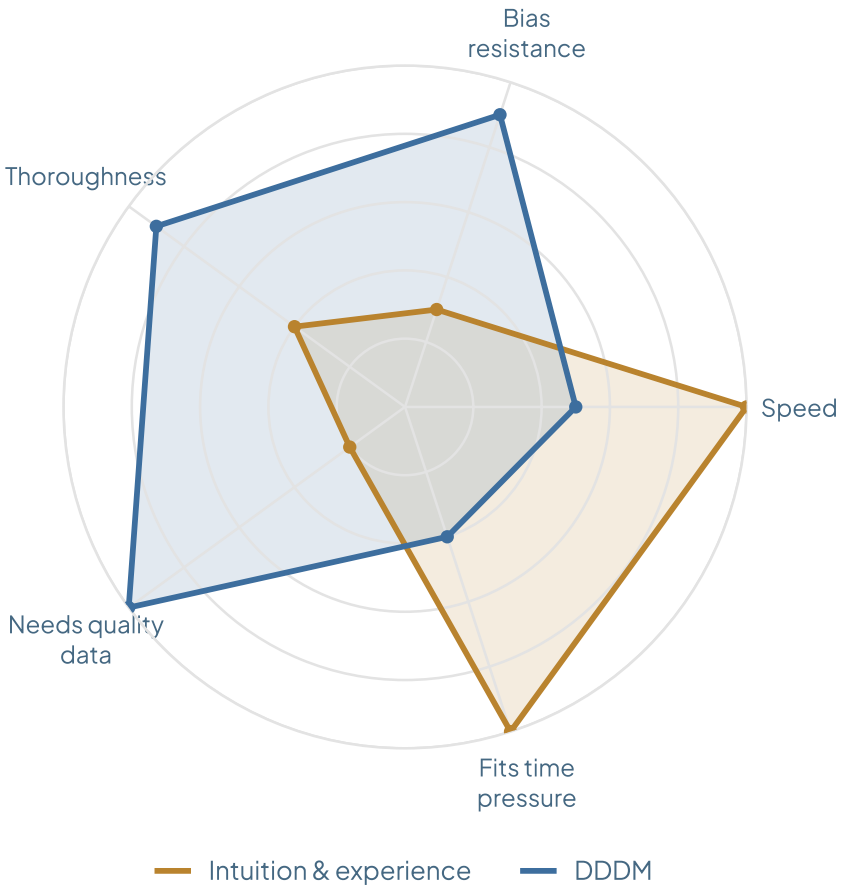
Strategic, tactical, operational decisions; five stages. Traditional (intuition/experience) vs DDDM (validate before acting: surveys, user tests, test markets). Data and intuition form a feedback loop.

2 Intuition and data as partners Deep dive

Intuition asks → Data tests → Results refine intuition

→ Intuition flags bad data

Traditional vs data-driven (conceptual)

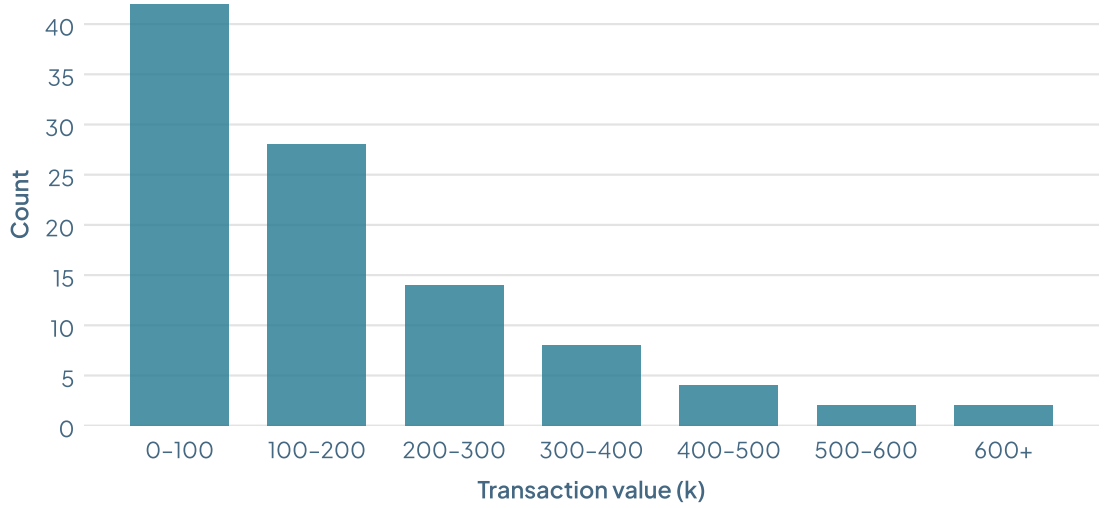


3 Descriptive analytics and statistics

Descriptive summarises what happened; diagnostic asks why. Centre, spread, shape, relationships. Zara uses all four levels for fast fashion supply.

4 Descriptive statistics that belong together Deep dive

A right-skewed distribution (illustrative)



Mode 0–100, median ~130, mean ~180.

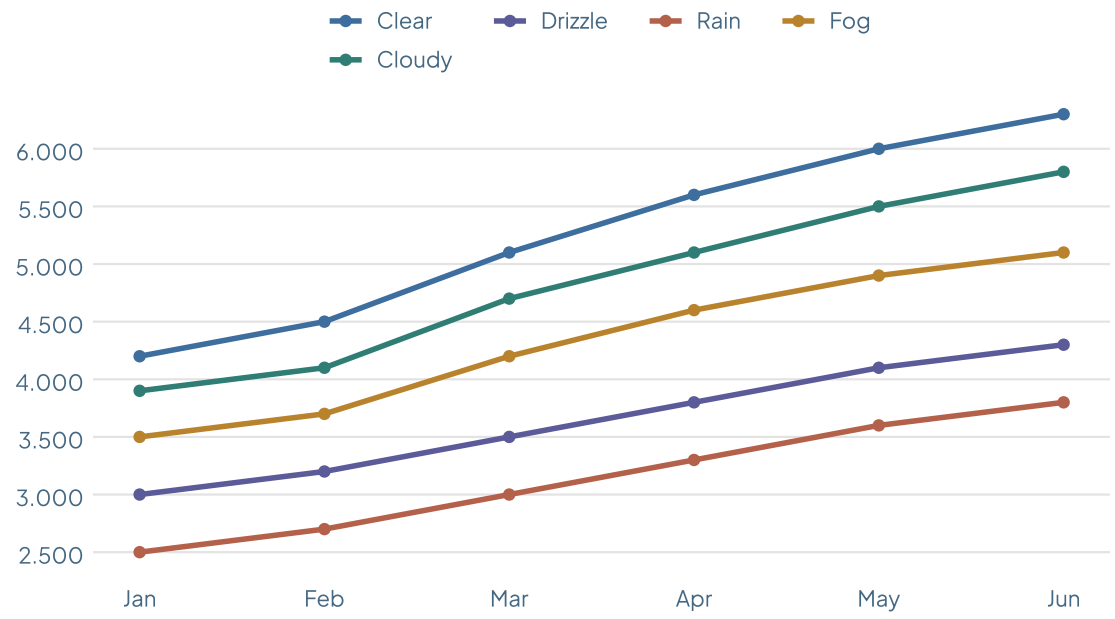
Right-skewed: mode < median < mean.

5 Visualisation and storytelling

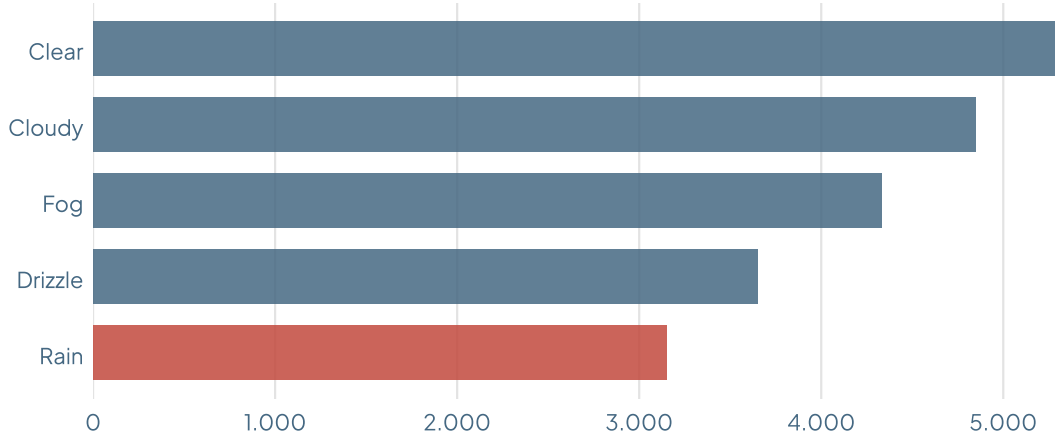
Choose charts by message (bar, line, stacked, histogram, scatter, map); avoid 3D, truncated bars, rainbow colours. Storytelling: context → finding → action; stories are ~22× more memorable. Declutter and use conclusion titles.

6 Before and after: from cluttered to a story Deep dive

Before: 'Bike rentals by weather'



After: 'Rainy days cut rentals ~40%'



Conclusion title, one highlighted category (illustrative data).

7 Descriptive statistics in one example, and why you must look at the chart

Example: delivery times (days) for 10 BetonPrima cement orders — **2, 3, 3, 3, 4, 4, 5, 6, 7, 15**.

Measure	How it is computed	Result
Mean	52 ÷ 10	5.2 days
Median	Average of the 5th and 6th values (4 and 4)	4 days
Mode	Most frequent value	3 days
Standard deviation (sample)	$\sqrt{[\Sigma(x - \text{mean})^2 \div (n - 1)]}$	3.77 days
Quartiles 1 and 3	25th and 75th percentiles	Q1 = 3; Q3 = 5.75
IQR	Q3 − Q1	2.75
Upper outlier fence	Q3 + 1.5 × IQR = 5.75 + 4.125	9.875 → the value 15 is an **outlier**

One order delivered 15 days late pulls the mean to 5.2, while half the orders arrive in ≤ 4 days. Without that outlier the mean drops to 4.1 and the SD from 3.77 to 1.62. **Do not delete outliers straight away** — first ask why. At BetonPrima, that 15-day order may be exactly the one during heavy rain: the outlier is what explains why customers leave.

An honest report

"Median delivery time is 4 days (IQR 3–5.75), but 1 in 10 orders was up to 15 days late." This sentence is far more useful to the board than "average 5.2 days".

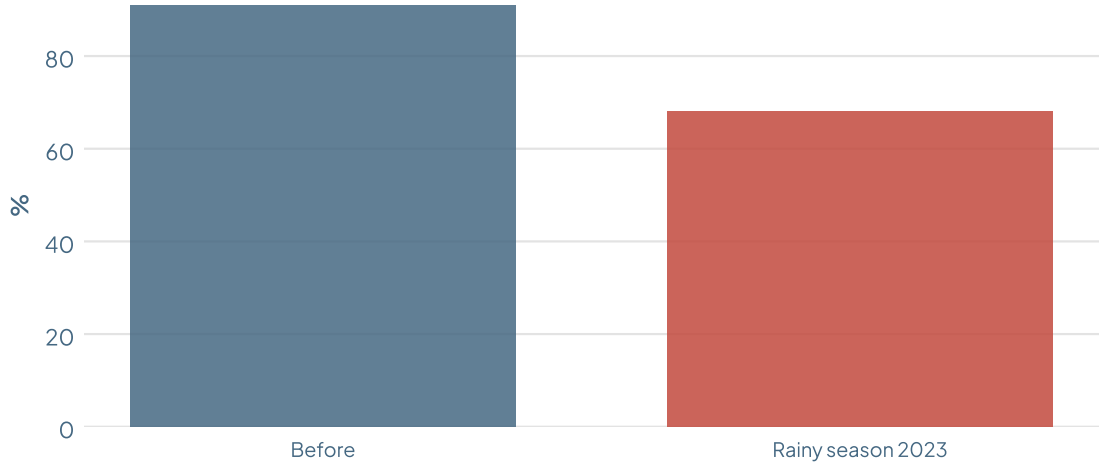
Anscombe's quartet (1973) is four data sets with almost identical means, variances, correlation (about 0.82) and regression lines — but when plotted they look completely different: one sensible linear pattern, one curve, one perfect line with a single outlier, and one nearly vertical cloud where one extreme point creates the correlation. The lesson: *summary statistics without a chart can mislead*. Always plot the data before concluding.

Diagnostic analytics techniques (answering "why?"): *drill-down* (from the national total to province, depot and customer), *segment comparison* (rainy vs dry days), *variance analysis* (budget vs actual: due to price, volume or mix?), *correlation and cross-tabulation*, and *root-cause analysis* (5 Whys, fishbone diagram). Remember: correlation points to candidate causes, not proof.

8 For BetonPrima: two charts that convince the board

Deep dive

West Java on-time delivery fell 23 points in the rainy season



Order-to-site rose to 5.4 days (target 3.0).

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Context → finding → action
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Key takeaways

1. Decision types.
2. DDDM validates first.
3. Data and intuition partner.
4. Describe centre, spread, shape.
5. Right chart, no clutter.
6. Context → finding → action.

Case studies

Case & worked answers

Bike sharing: from cluttered chart to story

Redesign the bike-sharing chart: one message, conclusion title, highlight rain, end with an action. Apply to BetonPrima: on-time decline line with rainy-season annotation plus churn-reason bars.

Worked examples

Problem 1 · Medium · **Descriptive statistics and outliers**

Customer waiting times (minutes) for 10 transactions: 4, 5, 5, 6, 6, 7, 7, 8, 9, 21. Compute the mean, median, mode, sample standard deviation, Q1, Q3, IQR, and check for outliers with the 1.5 × IQR rule.

1. Mean = 78 ÷ 10 = **7.8**. Median = average of the 5th and 6th values = (6 + 7) ÷ 2 = **6.5**. Mode: 5, 6 and 7 each appear twice (multimodal).

2. $s = \sqrt{[\Sigma(x - \bar{x})^2 \div (n - 1)]} \approx 4.87$ minutes

3. Q1 = 5.25; Q3 = 7.75 (Excel/NumPy interpolation). IQR = 7.75 – 5.25 = 2.5.

4. Upper fence = Q3 + 1.5 × IQR = 7.75 + 3.75 = 11.5 → **21 is an outlier**. Lower fence 5.25 – 3.75 = 1.5 → no low outliers.

5. Mean (7.8) > median (6.5) because the data is right-skewed. Report the median + IQR, and investigate the 21–minute transaction.

Answer: Mean 7.8; median 6.5; s ≈ 4.87; IQR 2.5; 21 minutes = outlier.

Problem 2 Challenging **Pearson correlation**

Promotion spend (x, IDR million) and sales (y, hundreds of units) over five weeks: (1, 2), (2, 4), (3, 5), (4, 4), (5, 5). Compute the correlation coefficient r and interpret it.

1. $\bar{x} = 3$; $\bar{y} = 20 \div 5 = 4$.
2. x deviations: −2, −1, 0, 1, 2. y deviations: −2, 0, 1, 0, 1. $\Sigma(dx \cdot dy) = 4 + 0 + 0 + 0 + 2 = 6$.
3. $\Sigma dx^2 = 10$; $\Sigma dy^2 = 4 + 0 + 1 + 0 + 1 = 6$.
4. $$r = \Sigma(dx \cdot dy) \div \sqrt{(\Sigma dx^2 \times \Sigma dy^2)} = 6 \div \sqrt{60} \approx 0.77$$
5. A fairly strong positive relationship. But there are only 5 points, and correlation ≠ causation — promotions may simply have run in busy weeks. Plot the data and test with an experiment.

Answer: $r \approx 0.77$.

Practice questions

Try answering first. The answer key is below each question.

1. Best chart for a time trend…

A. Pie
B. Line
C. Scatter
D. Map

Answer: B. Line.

2. A good storytelling title…

A. 'Rental chart'
B. 'Rain cuts rentals 40%'
C. '2024 data'
D. None

Answer: B. Title = conclusion.

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3. In right-skewed data, usually…

A. Mean < median < mode
B. Mode < median < mean
C. All equal
D. Median < mode < mean

Answer: B. The tail pulls the mean.

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4. Zara's fast allocation is…

A. Descriptive
B. Diagnostic
C. Predictive
D. Prescriptive

Answer: D. Decides action.

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5. DDDM needs all EXCEPT…

A. Clear goals
B. Quality data
C. Interpretation skills
D. Removing intuition entirely

Answer: D. They're partners.

Sources

- Week 3 MK6 Material — Descriptive and Diagnostic Analysis
- Knaflic, Storytelling with Data